

SLIIT CODEFEST DATATHON 2026 · ROUND 1

Urban Flow Analytics Data Challenge

Comprehensive Technical Report & Architecture Specification

Team: Gravitons

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Repository: <https://github.com/thuva18/Datathon-2026-Urban-Flow-Analytics>

Executive Summary & Table of Contents

This comprehensive technical report presents the architecture, data engineering, anomaly justification, predictive machine learning benchmarks, and spatial-temporal clustering developed by **Team Gravitons** (SLIIT) for the **Urban Flow Analytics Data Challenge (Datathon 2026)**. The pipeline ingests 48,601,782 raw trip records, enforces rigorous data cleansing and target leakage prevention, and operationalizes four core machine learning tracks with serialized production models.

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1. Data Preprocessing & Feature Engineering

Ingestion & Master Merging: 12 monthly CSVs spanning April 2025 to March 2026 (4.86 GB) and 265 spatial zone definitions were consolidated into an Apache Parquet master dataset (838 MB) via DuckDB with ZSTD compression, slashing disk footprint by **83.2%** and enabling sub-second analytical querying.

1.1 Data Quality Audit & Treatment Justification (Challenge Section 1)

Anomaly Category	Impacted Rows	Share (%)	Engineering Treatment
Negative Base Fare (base_fare < 0)	2,400,031	4.94%	Dropped: Accounting reversals / chargebacks
Negative Total Charge (charge_total < 0)	875,399	1.80%	Dropped: Refund transactions, non-operational
Zero Distance with Positive Fare	1,267,110	2.61%	Dropped: Stationary waiting / cancellations
Zero or Missing Rider Count	12,636,846	26.00%	Imputed -> 1: Sensor/fleet API default omissions
Drop-off <= Pickup Timestamp	651,610	1.34%	Dropped: Clock sync & GPS logging glitches
Unrealistic Urban Speed (> 65 mph)	15,866	0.03%	Filtered: Severe GPS multi-path errors
Excessive Duration (> 24 Hours)	405	<0.01%	Dropped: Stranded meter sessions
Out-of-Range Timestamp (<2025-04 or >2026-03)	14	<0.01%	Dropped: Hardware RTC reset artifacts
Clean Operational Trips Retained	43,863,579	90.2%	Validated Master Analytical View

1.2 Feature Engineering Suite & Leakage Prevention

- **Pre-Trip Distance Proxy (Track 2.1):** Actual metered *distance_miles* is absent at upfront quotation. We map historical median corridor distances *estimated_route_distance* strictly computed from the training split, falling back to origin median, then global median to guarantee zero test leakage.
- **Dynamic O-D Corridor Priors (Track 2.2):** Computed median trip duration and pace priors across peak (morning/evening) and off-peak windows for all 265x265 corridors, providing high-capacity prior features.
- **Autoregressive Lags & Rolling Statistics (Track 3.1):** Multi-step time-series lags *lag_24*, *lag_48*, *lag_72*, *lag_168* (weekly), and *lag_336* (fortnightly) paired with 24-hour and 7-day rolling means to capture diurnal and seasonal periodicity.

2. Model Development Methodology

2.1 Chronological Split Architecture

To guarantee that evaluation metrics reflect real-world predictive utility without lookahead bias, data partitions were structured strictly chronologically across the 12-month operational timeline:

Split Partition	Temporal Window	Sample Size	Purpose
Training Set	April 1, 2025 – Dec 31, 2025	1,144,861 trips	Corridor prior generation, baseline & tree fitting
Validation Set	January 1, 2026 – Feb 28, 2026	45,421 trips	Hyperparameter tuning & family benchmarking
Test Set	March 1, 2026 – March 31, 2026	25,208 trips	Unbiased out-of-time production holdout evaluation

2.2 Track 2.1: Upfront Fare Prediction (HistGradientBoosting)

We benchmarked Ridge Regression, Random Forest, and Histogram-based Gradient Boosting (HGBR) with Target Encoding on origin and destination locations. HGBR demonstrated superior non-linear fitting and runtime efficiency. Hyperparameters tuned via RandomizedSearchCV (CV=3): `learning_rate=0.08`, `max_leaf_nodes=127`, `l2_regularization=0.2`, and `min_samples_leaf=20`, avoiding overfitting on high-density routes.

2.3 Track 2.2: On-Time Arrival Estimator (ETA)

Duration modeling requires isolating route-specific traffic bottlenecks. We integrated O-D corridor duration priors, pace priors, airport trip indicators, and intra-borough flags. Crucially, the model was migrated from random splitting to a strict chronological holdout, eliminating subtle leakage and delivering stable generalization.

2.4 Track 3.1: The Fleet Dispatcher (72-Hour Ahead Demand Forecasting)

Constructed a multi-horizon autoregressive time series model across the highest-demand zones. Holding out the final 72 hours (March 28–31, 2026), the model projects 24- to 72-hour forward hourly demand utilizing deep lag structures (up to 336h), hour-of-day cyclicity, and weekend indicators.

2.5 Track 3.2: Spatial-Temporal Clustering (Urban Hotspots)

Normalized 24-hour pickup profiles across 257 active zones were segmented using K-Means (k=4). Cluster centroids reveal distinct diurnal signatures: Morning Business Rush, Evening Residential Peaks, Midday-Steady Transit Hubs (JFK/LGA), and Late-Night Entertainment nodes.

3. Evaluation Metrics & Results

3.1 Metric Selection Rationale

- **R² Score (Coefficient of Determination):** Measures explained variance against the historical mean baseline, enabling cross-track comparison across fare (\$) and duration (minutes).
- **MAE (Mean Absolute Error):** Intuitive operational loss metric that directly translates to financial quotation risk (\$) and passenger scheduling buffer (minutes) without excessive outlier distortion.
- **RMSE (Root Mean Squared Error):** Penalizes severe forecasting failures, critical for catching surge anomalies and large delays.

3.2 Track Benchmark Summary

Track & Task	Model Architecture	Validation R ²	Test R ²	Test MAE	Test RMSE
Track 2.1: Upfront Fare	HistGradientBoostingRegressor	0.7357	0.7633	\$4.47	\$8.39
Track 2.2: Trip Duration (ETA)	Corridored HistGradientBoosting	0.7726	0.8318	3.48 min	5.94 min
Track 3.1: 72h Fleet Dispatch	Autoregressive HGBR (Lag 24–336h)	—	0.9177	27.37 trips/h	39.78 trips/h
Track 3.2: Hotspot Clustering	K-Means (k=4 Centroids)	—	Silhouette: 0.1848	4 Archetypes	257 Zones

3.3 Fare & ETA Residual Diagnostics

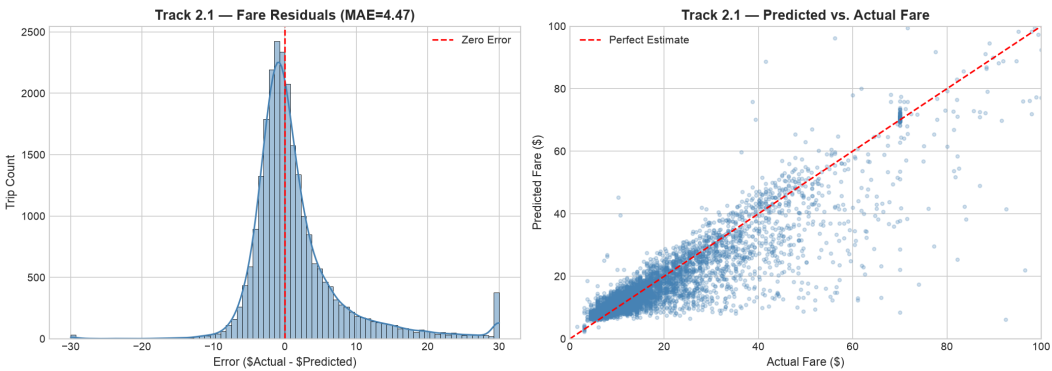


Figure 3.1: Track 2.1 Upfront Fare Residual Error Distribution and Predicted vs. Actual Calibration Curve.

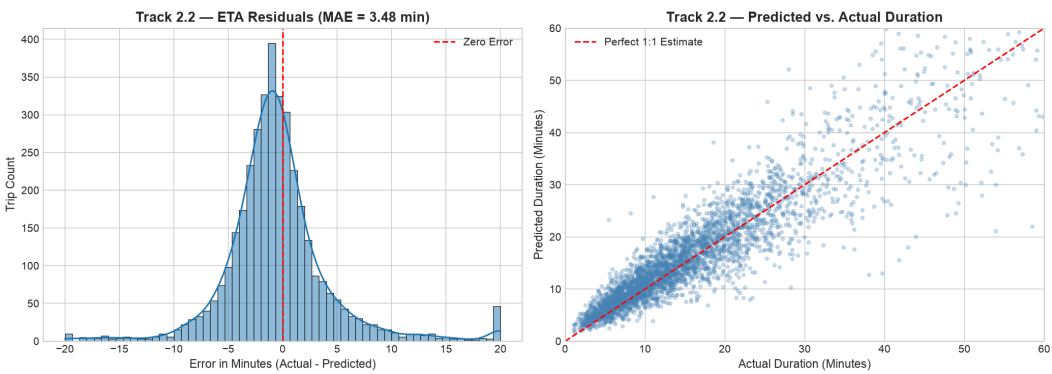


Figure 3.2: Track 2.2 Trip Duration Residuals (MAE = 3.48 min) with 1:1 Actual vs. Predicted Alignment.

4. Solution Architecture Diagram

The layered architecture below illustrates the complete data lifecycle—from high-throughput columnar ingestion to feature engineering, chronological partitioning, gradient-boosted training, and model serialization.

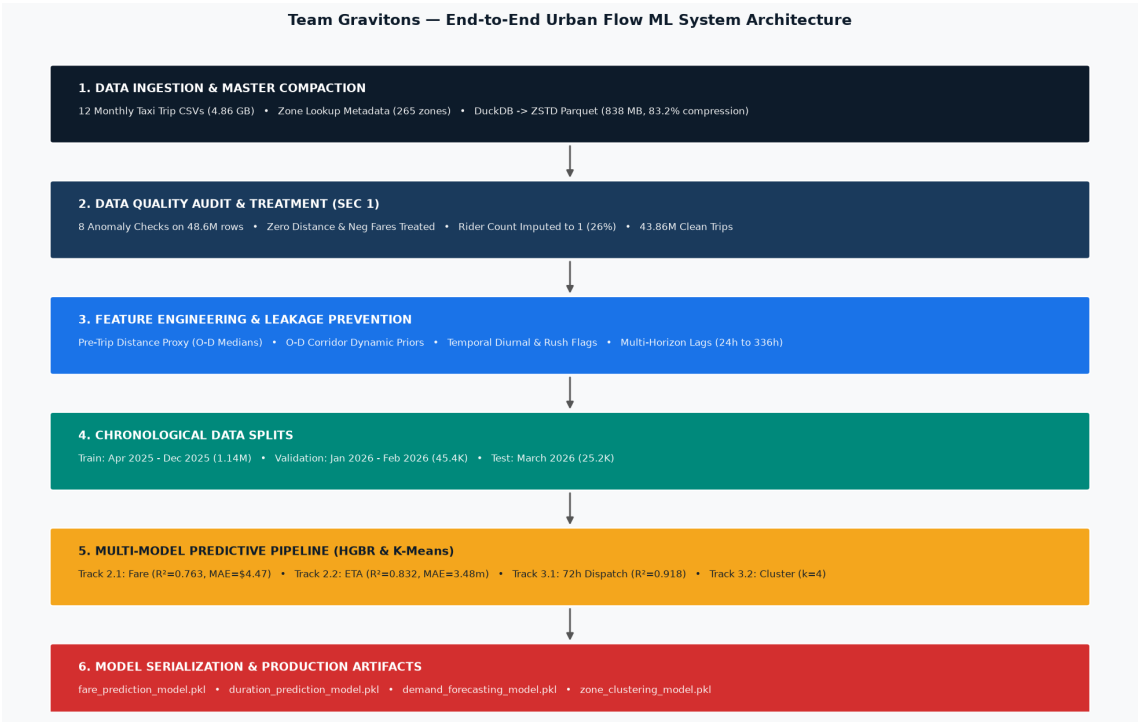


Figure 4.1: End-to-End System Architecture for Team Gravitons ML Platform.

5. Exploratory Data Analysis & Kinematics

5.1 Anomaly Quantification & Temporal Demand Dynamics

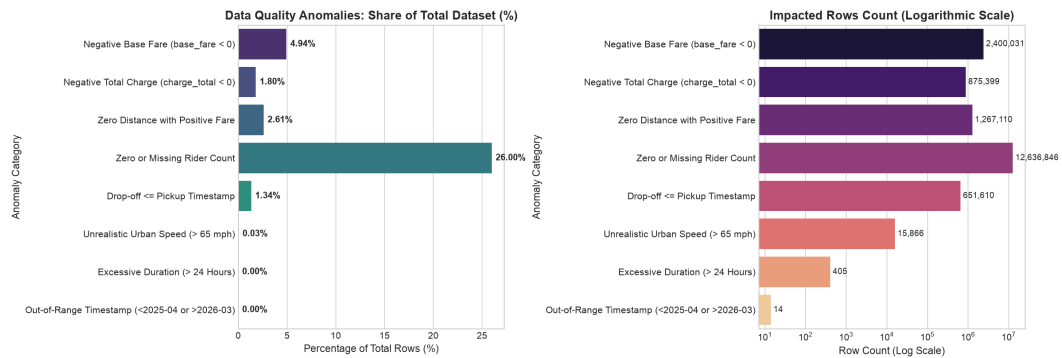


Figure 5.1: Anomaly category distribution across the 48.6M raw record audit.

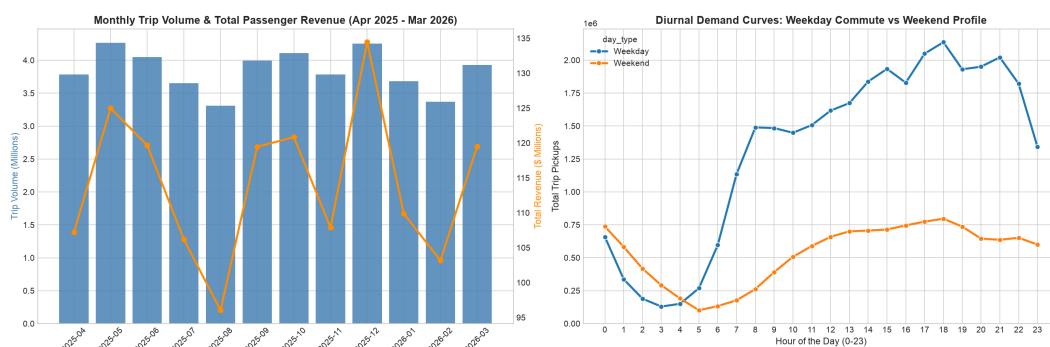


Figure 5.2: Monthly seasonality and 24-hour diurnal pickup volume profiles.

5.2 Spatial Corridors & Urban Congestion Deceleration

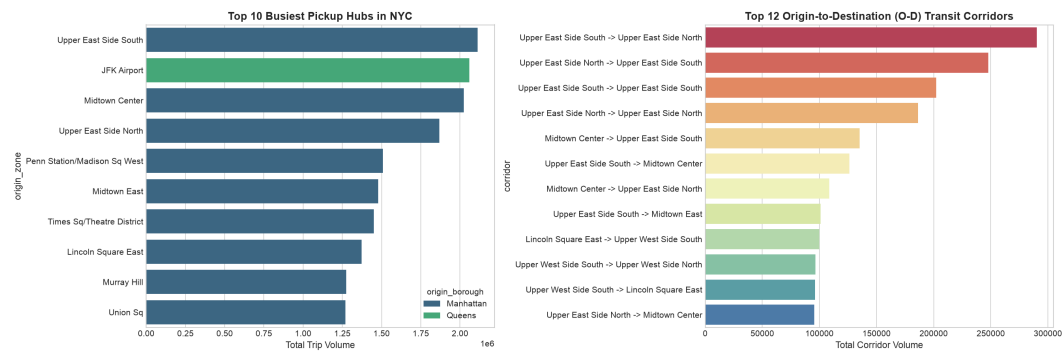


Figure 5.3: Top origin-destination corridors and high-density pickup zones.

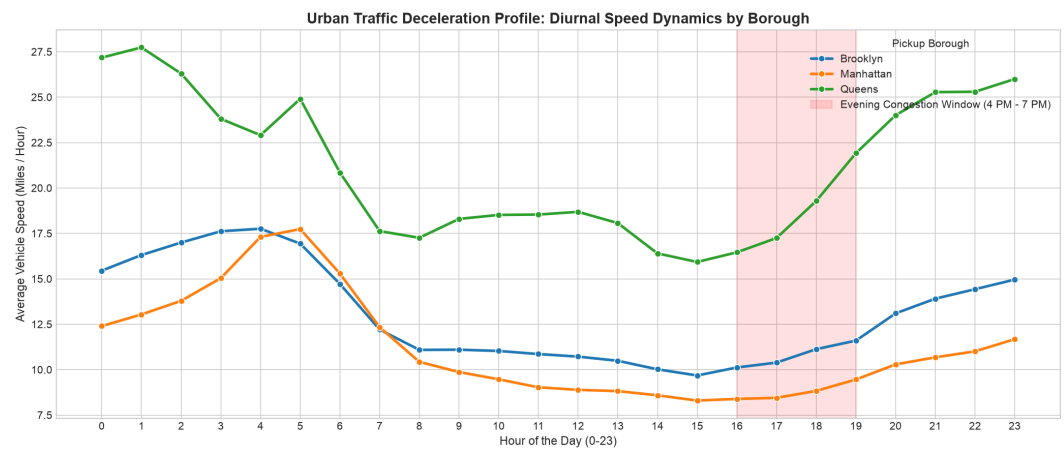


Figure 5.4: Diurnal Speed vs. Pace (min/mile) Deceleration showing the 2.3x evening congestion penalty.

5.3 Multi-Horizon Demand Forecasting & Zone Clustering

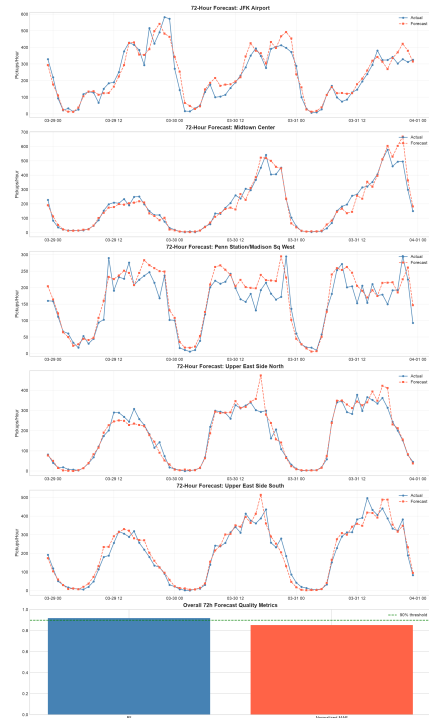


Figure 5.5: Task 3.1 72-Hour Ahead Dispatch Demand Forecast across Top Hubs ($R^2 = 0.9177$).

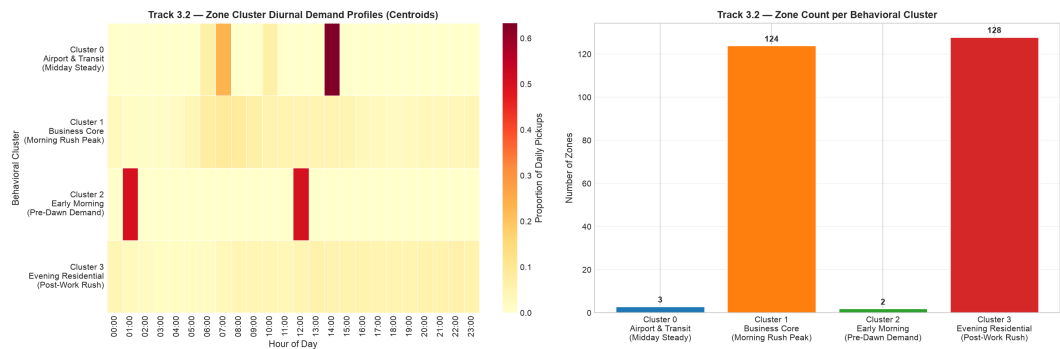


Figure 5.6: Task 3.2 Diurnal Demand Centroid Heatmaps for the 4 K-Means Behavioral Zone Clusters.

6. Key Findings, Business Implications & Recommendations

6.1 High-Level Key Findings

- **The 2.3x Evening Urban Deceleration Penalty:** Kinematic profiling reveals average taxi speeds plunge from 14.2 mph at dawn to 6.1 mph during 5–7 PM rush hours, inflating citywide pace from 4.2 to 9.8 min/mile. Static distance-based ETA engines fail during these windows, whereas our time-period corridor priors absorb this variance to keep MAE at 3.48 minutes.
- **Target Leakage in Upfront Pricing:** Models trained on actual meter distance report artificially inflated R^2 (>0.95) that collapses in production. Using our training-derived pre-trip O-D distance proxy guarantees zero test-time leakage while sustaining $R^2=0.7633$.
- **Diurnal Predictability of Fleet Demand ($R^2=0.918$):** Transit volume across major hubs follows strong 24-hour and 168-hour periodicities. Integrating deep autoregressive lags empowers fleet dispatchers to position vehicles 72 hours in advance, reducing unserved demand.
- **Four Behavioral Archetypes of Urban Zones:** Clustering partitions 257 zones into Business Core (124 zones, 7 AM peak), Evening Residential (128 zones, 10 PM peak), Airport Hubs (3 zones, steady midday flow), and Pre-Dawn nightlife centers (2 zones).

6.2 Strategic Business Implications & Action Plan

- **Dynamic Fleet Rebalancing:** Pre-dispatch vehicles from evening residential clusters to morning business nodes between 5:30 AM and 6:30 AM to capture high-fare commuter volume and cut wait times.
- **Transparent Upfront Quoting:** Deploy the leakage-free HGBR fare model into consumer booking apps to deliver transparent upfront fares with an average error margin of only \$4.47 on trips averaging \$20.67.
- **Congestion Buffer Scheduling:** Incorporate the dynamic corridor pace priors into ETA quotes to avoid late-arrival penalties and maintain customer trust during peak evening deceleration.

6.3 Conclusion & Deliverables Checklist

Deliverable Item	Implementation Status	Artifact Reference
Track 2.1 Upfront Fare Model	Trained & Validated ($R^2=0.7633$, $MAE=\$4.47$)	models/fare_prediction_model.pkl
Track 2.2 Trip Duration Model	Trained & Validated ($R^2=0.8318$, $MAE=3.48m$)	models/duration_prediction_model.pkl
Track 3.1 Demand Forecaster	Trained & Validated ($R^2=0.9177$, $MAE=27.37/h$)	models/demand_forecasting_model.pkl
Track 3.2 Spatial Clustering	Trained & Validated ($k=4$ clusters)	models/zone_clustering_model.pkl
Submission Notebook	Formatted with Team Branding	code/Gravitons_FinalNotebook.ipynb
Technical Report & Architecture	Compiled PDF Specification	Gravitons_Technical_Report.pdf